

Shiva Upadhye

Curriculum Vitae

CONTACT INFORMATION

Shiva.Upadhye@tufts.edu
Psychology Building, Tufts University
490 Boston Ave, Medford MA 02155

Appointments

Postdoctoral Scholar
Department of Psychology, Tufts University
NeuroCognition of Language Lab

August 2026 - Present

Education

University of California, Irvine

Ph.D. in Language Science

June 2026

Dissertation: *Information and Performance Constraints in Online Language Production*

Committee: Richard Futrell (chair), Gregory Hickok, Gregory Scontras, Xin Xie, and Victor Ferreira

University of California, San Diego

Bachelor of Science in Cognitive Science

June 2020

Concentration: Machine Learning and Neural Computation

Minor: Mathematics

Publications

Upadhye S. & Futrell, R. Back to the Future: The Role of Past and Future Context Predictability in Incremental Language Production. *Open Mind. (In press)*.

Baidildinova M., **Upadhye S.**, Wagner A., Mayer M., & Futrell, R. (2025). Quantifying Mutual Intelligibility using Language Models: Evidence from Turkic Languages. *Proceedings of the Society of Computation in Linguistics*.

Upadhye S. Li Y., Li J. Attali, N. & Scontras G. (2025). GPT-2's Production and Comprehension of Scope Ambiguity (2025). *PragLM @ Conference on Language Modeling*.

Upadhye S. & Futrell, R. (2025). Examining Future Context Predictability Effects in Word-form Variation and Word Choice. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society*.

Upadhye S., Li J., & Futrell, R. (2025). SPACER: A Parallel Dataset of Speech Production And Comprehension of Error Repairs. *The 14th edition of the Workshop on Cognitive Modeling and Computational Linguistics*.

Upadhye S. & Futrell, R. (2023). Typing time PWI: A scalable paradigm for studying lexical production. *Proceedings of the 45th Annual Meeting of the Cognitive Science Society*.

Upadhye, S. Li, J., & Futrell, R. (2023) Bridging production and comprehension: A computational model of error generation and correction. *Proceedings of the Society for Computation in Linguistics*.

Upadhye S. & Futrell, R. (2022). Information-theoretic analyses of disfluencies in Speech. *NeurIPS 2022 Workshop on Information-Theoretic Principles in Cognition Systems*.

Upadhye S., Bergen L., & Kehler, A (2020). Predicting Reference: What do Language Models learn about Discourse Models? *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*.

**Conference
Presentations**

Upadhye S. & Futrell, R. (2026). Repairing, Fast and Slow: Lexical and Contextual Pressures on the Timecourse of Error Processing in Naturalistic Production. *The 39th Annual Conference on Human Sentence Processing*. Cambridge, MA, USA. (**Poster**).

Baidildinova, M., **Upadhye S.**, Wagner, A., Mayer, C., & Futrell, R. (2026). Quantifying Mutual Intelligibility using Language Models: Evidence from Turkic Languages. *The 39th Annual Conference on Human Sentence Processing*. Cambridge, MA, USA. (**Poster**).

Upadhye S., Li Y., Li J. Attali, N. & Scontras G. (2025). GPT-2's Production and Comprehension of Scope Ambiguity (2025). *PragLM @ Conference on Language Modeling*. Montreal, Quebec, Canada. (**Poster**)

Upadhye S., Yeaton, JD., Yi, E., & Hickok, G. (2025). Speakers and typers adapt their planning strategies to meet sentence production demands. *66th Annual Meeting of the Psychonomics Society*. Denver, CO, USA. (**Poster**)

Upadhye S. & Futrell, R. (2025). Examining Future Context Predictability Effects in Word-form Variation and Word Choice (2025). *47th Annual Meeting of the Cognitive Science Society*. San Francisco, CA, USA. (**Talk**)

Upadhye S., Yeaton, JD., Yi, E., & Hickok, G. (2025). Typing Fluency reveals Psycholinguistic Dynamics of Sentence Production. *The 38th Annual Conference on Human Sentence Processing*. College Park, MD, USA. (**Poster**)

Li J., **Upadhye S.**, Gandhasri, R., & Futrell, R. (2025). SPACER: A Parallel Dataset of Speech Production And Comprehension of Error Repairs. *The 38th Annual Conference on Human Sentence Processing*. College Park, MD, USA. (**Poster**)

Li Y., Li J., **Upadhye S.**, & Scontras, G. (2025). Language models prefer ambiguous utterances following informative contexts. *The 38th Annual Conference on Human Sentence Processing*. College Park, MD, USA. (**Poster**)

Yeaton, JD, **Upadhye, S.**, Futrell, R., & Hickok, G. (2024). The (Non-)Sentence Repetition Task: A novel assessment of expressive and receptive syntactic competence. *16th Annual Meeting of the Society for the Neurobiology of Language*. Brisbane, Australia. (**Poster**)

Upadhye S., Attali N. & Scontras, G. (2023). The role of world knowledge for interpreting aspect ambiguity. *Tenth Biennial Meeting of Experimental Pragmatics*. Paris, France. (**Poster**)

Li, J., **Upadhye, S.** & Futrell, R. (2023) Bridging production and comprehension: A computational model of error generation and correction. *The 36th Annual Conference on Human Sentence Processing*. Pittsburgh, PA, USA. (**Poster**)

	Upadhye S. & Futrell R. (2022). Information-theoretic analysis of disfluencies in speech. <i>NeurIPS 2022 Workshop on Information Theoretic Principles in Cognitive Systems</i>. New Orleans, LA, USA. (Poster)	
Other Contributions	Moslehi B., Price W., Moslehi B., Off J., Sourichanh K., Cheng Y., Upadhye S., & Soutoudeh V. (2025). <i>Dual cavity interferometer sensor system</i>. U.S. Patent No. US20250224295A1. US Patent & Trademark Office.	
Teaching & Mentorship	UC Irvine Instructor of Record Introduction to Linguistics	Summer 2025
	Pedagogical Training Course Development Fellowship for Language Science (2024) UCI Division of Teaching Excellence & Innovation - Graduate Summer Fellowship & Certificate for Hybrid Inclusive Pedagogy (2022)	
	Teaching Assistant Introduction to Linguistics, Research Methods in Psychology, Psychology of Language, American Public Policy, Acquisition of Language, Discovering Language;	2022 - 2025
	Mentorship Rutvik Mohan Gandhasri, Mackenzie Leigh Maxwell (UROP Fellow - 2024), Frankie Boren, Austin Wagner (UROP Fellow - 2025)	2023 - Current
	UC San Diego Instructional Assistant Computational Linguistics	Spring 2020
Service, Fellowships, & Awards	Program Chair and Organizer Society for Computation in Linguistics (SCiL 2024)	
	Reviewing Communications Psychology; Annual Meeting of the Cognitive Science Society (Cogsci); BlackboxNLP; Society for Computation in Linguistics (SCiL); California Meeting on Psycholinguistics (CAMP)	
	Awards & Fellowships Associate Graduate Dean's Fellowship, UC Irvine (2021 - 2026; 3 quarters)	
	UCI Center for Theoretical Behavioral Sciences (CTBS) Travel Award (2025)	
	Simmons Institute Analytical Connectionism Summer School Fellowship (2024; declined)	
	Intel Academic GPU Donation Program (2023) - Intel Arc A770 GPU	

**Other
Experience**

AI & Data Analytics Engineer
Intelligent Fiber Optic Systems Corporation

Oct 2020 - July 2021

Computational Neurosciences Intern
National Center for Microscopy and Imaging Research

Oct 2019 - Mar 2020

Research Assistant & Undergraduate Researcher
Computational Linguistics Lab (UC San Diego)

Mar 2019 - June 2020